

Build a student grade tracker using a list of dicts — each dict holds name, scores (list), and computed average; sort the list by average descending and print the leaderboard.

```
students = [
    {"name": "Ana", "scores": [90, 85, 88]},
    {"name": "Maria", "scores": [75, 80, 78]},
    {"name": "Camelia", "scores": [95, 92, 90]}
]
for student in students:
    average = sum(student["scores"]) / len(student["scores"])
    student["average"] = average

students.sort(key=lambda student: student["average"], reverse=True)
print("Leaderboard")
for position, student in enumerate(students, start=1):
    print(f"{position}. {student['name']} - Average: {student['average']:.2f}")
```

```
Leaderboard
1. Camelia - Average: 92.33
2. Ana - Average: 87.67
3. Maria - Average: 77.67
```

Explanation: I've created a list of dictionaries, where each dictionary stores a student's name and a list of scores. It calculates the average score for each student and saves it in the dictionary. The list is then sorted by the average score in descending order (highest to lowest). In the end, the program prints a leaderboard showing each student's rank, name, and average score.

Exercise 2: Write a list comprehension that filters even numbers from a range and squares them, then write an equivalent dict comprehension mapping each word in a sentence to its length.

```
even_squares = [number ** 2 for number in range(1, 11)
                 if number % 2 == 0]

print(even_squares)
```

```
[4, 16, 36, 64, 100]
```

Explanatuion: I've made a program that uses a list comprehension to select only the even numbers from 1 to 10 and stores their squares in a list.

Exercise 3: Read a CSV file with the csv module, parse every row into a dict, compute column averages, and write a summary CSV with the results.

```
import csv

total_math = 0
total_english = 0
count = 0
with open("students.csv", "r") as file:
    reader = csv.DictReader(file)
    for row in reader:
        total_math += int(row["Math"])
        total_english += int(row["English"])
        count += 1
average_math = total_math / count
average_english = total_english / count
with open("summary.csv", "w", newline="") as file:
    writer = csv.writer(file)

    writer.writerow(["Subject", "Average"])
    writer.writerow(["Math", average_math])
    writer.writerow(["English", average_english])

print("Summary CSV created successfully.")
```

Explanation: The program uses the csv module to read data from a CSV file. Each row is read as a dictionary using csv.DictReader(). It adds the Math and English marks, calculates the average for each subject, and writes the results to a new CSV file called summary.csv using csv.writer().

Exercise 4: Use the os and pathlib modules to walk a directory tree, list all .txt files, and print their sizes in a formatted table.

```
import os
from pathlib import Path
folder = Path(".")
print(f"{'File Name':<20} {'Size (bytes)':>12}")
print("-" * 35)
for root, dirs, files in os.walk(folder):
    for file in files:
        if file.endswith(".txt"):
            file_path = Path(root) / file
            size = file_path.stat().st_size
            print(f"{'file':<20} {'size':>12}")
```

The program is using the os and pathlib modules to search through a folder and its subfolders. It looks for files that end with .txt, gets the size of each file in bytes, and prints the file names and sizes in a formatted table